

INTACT: Intelligent Network Topology-Aware Collaborative Training for Cyber-Physical Intrusion Detection in DER Networks

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Abstract—As modern distribution networks integrate Distributed Energy Resources (DERs), the resulting bidirectional power flow creates conditions under which stealthy replay attacks evade localized detection. While Federated Learning (FL) offers a privacy-preserving solution, existing methods are fundamentally topology-blind, evaluating edge nodes in isolation without enforcing the underlying electrical constraints of the network. To address this, we propose INTACT (Intelligent Network Topology-Aware Collaborative Training), deployed on an IEEE 33-bus radial network partitioned by electrical proximity. Each zone trains a localized LSTM Autoencoder on Active Power, Reactive Power, Voltage Magnitude, and Phase Angle, then transmits model weights to a central server that aggregates contributions weighted by inter-zone electrical admittance. After each round, the server cross-validates the reconstruction residuals between neighboring zones. Because replay attacks corrupt only one zone while leaving neighbors unaffected, the resulting spatial mismatch exposes threats that isolated models cannot detect. Evaluated on a 4,002-window dataset against four FL baselines, INTACT achieves an F1-score of 0.730, outperforming FedAvg (F1=0.520) and falling within 7 percentage points of the centralized upper bound (F1=0.800), without compromising data privacy.

Index Terms—Anomaly Detection, Cyber-Physical Systems, Distributed Energy Resources, Federated Learning, Replay Attacks, Topology-Aware Networks.

I. INTRODUCTION

Historically, electrical distribution networks operated as passive one-way systems that delivered power from substations to consumers. However, the rapid growth of Distributed Energy Resources (DERs) has changed this structure. The grid now behaves as a dynamic bidirectional cyber-physical system (CPS) [1]. Although this shift enables decentralized energy management, it also introduces instability. Weather conditions such as sunlight, wind, and temperature directly influence power generation, causing continuous and unpredictable changes in power flow. This creates a noisy and complex operating environment, making it harder to separate normal behavior from attacks and ultimately increasing the cyber-physical attack surface.

This inherent variability creates an ideal environment for sophisticated cyberattacks to remain undetected. Although conventional False Data Injection Attacks (FDIAs) often focus on manipulating sensor values directly, more advanced temporal attacks such as replay attacks exploit the timing

and continuity of time-series data, making them significantly harder to detect within naturally volatile DER environments [2]. An attacker can record legitimate grid measurements during low-demand periods and replay them during peak demand to conceal malicious actions. Because the injected values remain within normal ranges, standard threshold-based detectors do not raise alarms. Instead, the attack takes advantage of the natural variability of DERs. From an isolated edge node's perspective, a replayed active power signal can be indistinguishable from natural fluctuations caused by weather variations such as cloud cover [3].

To secure decentralized grid infrastructure while preserving the privacy of sensitive operational data, researchers have increasingly adopted Federated Learning (FL) frameworks [4], [5]. However, existing FL-based intrusion detection methods remain largely topology-blind. They model DERs as independent entities and lack the spatial awareness needed to enforce physical constraints such as Kirchhoff's voltage and current laws across multi-node distribution networks [6]. As a result, temporally consistent but malicious signals can evade detection at the local level, even when they violate the physical consistency of the broader system.

To bridge the gap between IT-focused anomaly detection and OT-level physical constraint enforcement, this paper proposes the Intelligent Network Topology-Aware Collaborative Training (**INTACT**) framework. We utilized the IEEE 33-bus distribution system [7] to simulate a radial distribution network, specifically modeling the integration of active solar DERs to capture the complex dynamics of bidirectional power flow. The grid is partitioned into operational zones based on electrical proximity, where each zone is monitored by a localized edge computing client that trains a Long Short-Term Memory (LSTM) Autoencoder on its local physical state variables: Active Power (P), Reactive Power (Q), Voltage Magnitude ($|V|$) and Phase Angle (θ). Rather than centralizing raw grid data, zonal clients periodically transmit model weights to a federated aggregation server. Unlike standard Federated Averaging (FedAvg) [8], which treats all clients equally, our server assigns higher aggregation weight to zones that are more strongly connected in the physical network, measured by the admittance of the lines between them.

TABLE I: Comparison of Related Intrusion Detection Works

Method	Physical System Model	Attack Type	Key Limitation	Ref
Hybrid Federated Learning	None (communication layer only)	DoS & Data Integrity	Secures DNP3/Modbus protocols; no physical power flow modelling	[9]
Federated Deep Learning	Physical measurements (CPPS)	FDIAs	Does not quantify detection impact of DER-induced bidirectional flow	[10]
Vertical Federated Learning	Various sub-network measurements	FDIAs	No comparison of pre-DER vs. post-DER grid dynamics; unidirectional focus	[11]
Graph-Based Federated Learning	Power Distribution Systems	Natural Electrical Faults	Targets natural faults rather than cyberattacks; does not account for bidirectional power flow vulnerabilities from active solar integration	[12]
Physics-Informed LSTM	FOC-VSM swing equations	FDIAs on Control Layer	Limited to grid-forming VSM control; not applicable to radial DER distribution networks	[13]
Physics-Informed Conv. AE	DER-integrated grids (P , Q , V , θ)	Stealthy FDIAs	Centralized data dependency; incompatible with federated edge deployment	[14]
Adversarial Autoencoder	DER-integrated unbalanced grids	FDIAs	Centralized architecture; raw data pooling incompatible with edge privacy	[15]
Federated LSTM Autoencoder	Substation sensor & IoT data	Generic anomalies	No power flow variables; topology-blind; no temporal attack evaluation	[16]
INTACT	IEEE 33-bus radial (P, Q, V, θ) with solar DERs	Temporal (Replay)	Combines federated privacy, topology-aware aggregation, DER bidirectional flow modeling, and temporal attack resilience	Proposed

This ensures electrically coupled zones have greater influence on each other's learned models. After each aggregation round, the server cross validates reconstruction residuals between neighboring zones. Since real physical disturbances propagate across connected zones simultaneously, a replay attack corrupting only one zone is exposed through this spatial mismatch. Evaluated against four baselines on a 4,002-window dataset, INTACT achieves an F1-Score of 0.730. This outperforms FedAvg (F1=0.520) and closes 75 percent of the gap to the centralized upper bound (F1=0.800) while preserving full data privacy. The specific contributions of this paper are as follows:

- **DER-Integrated Grid Simulation:** Comparative characterization of a passive unidirectional baseline grid and an active DER-integrated network, quantifying how solar DERs reshape inter-zone coupling and create the operating conditions under which temporally consistent replay attacks evade localized detection.
- **Temporal Attack Modeling:** The synthetic injection of stealthy replay attacks into a multi-variable IEEE 33-bus dataset with active solar DERs, demonstrating that topology-blind models fail to reliably detect temporally consistent replay signals under natural DER variability.
- **Topology-Aware Federated Architecture:** An FL framework that weights client contributions based on inter-zone electrical admittance and uses cross-zone residual consistency checks to detect stealthy replay attacks, without sharing raw operational data.

The remainder of this paper is organized as follows. Section II reviews related work, Section III describes the simulation and attack methodology, Section IV presents the baseline models and edge testbed, Section V introduces INTACT, Section VI discusses the results, and Section VII concludes the paper.

II. RELATED WORK

Research on Intrusion Detection Systems (IDS) in cyber-physical power systems has generally developed in two distinct directions: FL approaches that prioritize communication privacy and physics-informed methods that leverage power flow constraints for anomaly detection. This section reviews the most relevant work along both areas and identifies the gap that motivates the proposed INTACT framework. Table I summarizes these studies, along with their key methods and limitations.

A. Federated Learning for Grid Intrusion Detection

Most FL approaches established the viability of distributed anomaly detection without centralizing the raw operational data. Purohit et al. [9] proposed FL-ADS, a federated anomaly detection system for DER networks targeting communication protocols such as DNP3 and Modbus. While the system effectively coordinates distributed clients, it secures the communication layer without modeling physical power flow variables, remaining blind to the physical coupling structure of the underlying network. Kausar et al. [10] developed a federated deep learning model using physical measurements for FDIA detection in cyber-physical power systems, yet do not isolate the vulnerability shift introduced when dynamic solar DERs are added to radial grids, leaving the impact of bidirectional power flow on detection performance unquantified. Kesici et al. [11] addressed data partitioning through vertical FL on the IEEE 123-bus and IEEE 37-node systems, achieving strong FDIA detection using various sub-network measurements. While the spatial-temporal split learning architecture represents a step toward physical awareness, the work focuses on static network partitioning and does not examine how the transition from

passive unidirectional grids to active DER-integrated networks reshapes the detection landscape.

Recent studies have attempted to incorporate spatial relationships into these architectures. Bandurin et al. [12] introduced a graph based FL approach specifically for fault detection in power distribution systems. However, their methodology targets the identification of natural electrical faults rather than sophisticated temporal cyberattacks, and it does not account for the bidirectional power flow vulnerabilities created by active solar integration.

B. Physics Informed Anomaly Detection

Physics informed approaches have improved intrusion detection by integrating power system constraints directly into learning models. Khaleghi et al. [13] combined control layer grid physics with a federated LSTM model to improve cyberattack detection in virtual synchronous machines. While effective and privacy preserving, this approach is limited to isolated grid forming control systems and does not capture the multi node spatial topology of radial distributed energy resource networks. Jabbari Zideh and Solanki [14] further improved physics aware intrusion detection using a multivariate convolutional autoencoder tested on the IEEE 123-bus system and a real-world distribution feeder. Although effective against stealthy FDIAs, the model relies on centralized data collection, limiting its suitability for federated edge environments.

C. Unsupervised Methods

Unsupervised learning has also been applied to integrated power systems. Jabbari Zideh et al. [15] developed an adversarial autoencoder that effectively models grid variability in unbalanced distribution networks. However, its centralized architecture requires raw data sharing, creating privacy and communication challenges that federated systems aim to eliminate. Most closely related to the architecture proposed here, Shrestha et al. [16] proposed a federated LSTM autoencoder for smart grid anomaly detection using substation sensor and IoT data at edge nodes. However, their model lacks physical power flow awareness, does not incorporate grid topology, and has not been evaluated against advanced temporal attacks such as replay.

D. Research Gap

Existing work shows that most federated methods preserve privacy but remain topology blind. While some recent physics informed approaches have adopted federated architectures for specific control systems, they do not scale to the physical constraints of multi node distribution networks. Furthermore, the majority of physics aware models that successfully capture network level behaviors still rely on centralized data. No existing approach combines federated privacy, topology awareness, bidirectional dynamics, and temporal attack detection in radial distribution networks.

III. GRID SIMULATION AND THREAT MODELING

A. IEEE 33-Bus Simulation Setup

The IEEE 33-bus radial distribution system [7] serves as the simulation testbed. This benchmark network features a single feeder originating from a main slack bus (Bus 0, fixed at 1.0 pu). Since Bus 0 represents the main substation, it is excluded from all local zone partitions. The feeder supplies 32 load buses in a radial topology, accurately representing a standard neighborhood distribution network. All simulations run on PandaPower, an open source power flow simulator [17].

To simulate DER integration, three solar photovoltaic units were placed at Bus 17 (0.5 MW), Bus 24 (0.4 MW), and Bus 32 (0.6 MW) across different network branches. Solar generation follows a bell-shaped diurnal profile at 5-minute intervals and operates only between 06:00 and 20:00. The simulation covers four weeks, generating 8,064 timesteps per bus. At each timestep, four key physical variables are recorded: Active Power (P), Reactive Power (Q), Voltage Magnitude ($|V|$), and Phase Angle (θ). Two simulation scenarios are generated under identical load profiles and weather conditions:

- **No-DER baseline:** DER output is set to zero at all three injection buses, representing a passive unidirectional distribution network.
- **DER-active scenario:** All three solar units operate normally, introducing bidirectional power flow during solar generation hours.

The grid is divided into four operational zones based on electrical proximity and feeder structure, as defined in Table II.

TABLE II: Zone Partitioning of the IEEE 33-bus System

Zone	Buses	DER Present
Zone 1	1–8	None
Zone 2	9–17	DER_1 at Bus 17 (0.5 MW)
Zone 3	18–24	DER_2 at Bus 24 (0.4 MW)
Zone 4	25–32	DER_3 at Bus 32 (0.6 MW)

To reflect realistic sensor measurement uncertainty, zero-mean Gaussian noise ($\sigma = 0.002$ pu) was added to both training and test data, consistent with typical distribution-level sensor accuracy.

B. Grid Comparison With & Without DER

To measure the impact of DER integration on the operating environment of anomaly detection systems, the two simulation scenarios are compared across voltage magnitude and active power dimensions. During the solar window, all three DER buses exhibit strongly negative mean active-power difference (ΔP), with peak injections of -0.6 MW (Bus 32), -0.5 MW (Bus 17), and -0.4 MW (Bus 24) at approximately 12:00. Outside the solar window, $\Delta P = 0$ across all buses, confirming identical nighttime behaviour.

The effect on the voltage magnitude distribution is shown per zone in Fig. 1. Statistical significance between the baseline and DER-integrated voltage distributions was evaluated using Welch's t -test [18]. DER raises mean voltage by $+14.1$ mpu

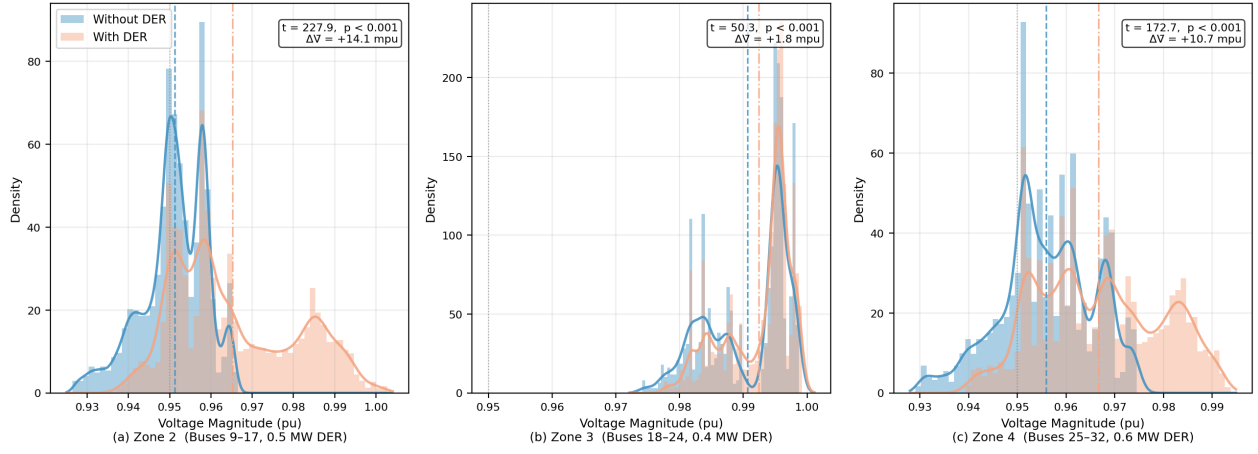


Fig. 1: Voltage magnitude distributions with and without DER integration.

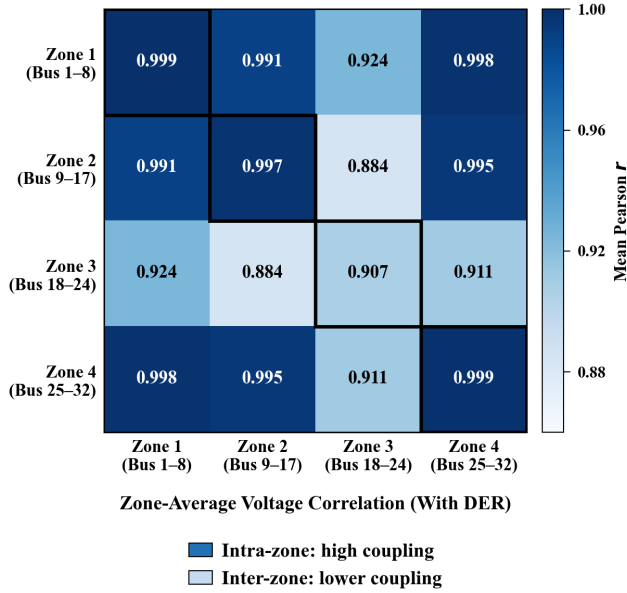


Fig. 2: Zone-average Pearson correlation of voltage magnitude under DER-active conditions. Bus 0 (slack bus) excluded.

in Zone 2 ($t = 227.9$, $p < 0.001$), +10.7 mpu in Zone 4 ($t = 172.7$), and only +1.8 mpu in Zone 3 ($t = 50.3$) despite its 0.4 MW capacity, reflecting greater electrical proximity to a well-regulated node. Across all DER-connected zones, voltage violations below 0.95 pu fall by 68.4% (from 44,012 to 13,920 bus-timesteps). This heterogeneous, zone-dependent shift elevates the reconstruction noise floor differently in each zone, motivating the per-zone LSTM autoencoders described in Section IV.

C. Inter-Zone Coupling & Topological Heterogeneity

To validate the physical motivation for topology-aware federated aggregation, Fig. 2 presents the zone-average Pearson correlation of voltage magnitude time series across all four zones under DER-active conditions.

Intra-zone correlations (diagonal) consistently exceed $r = 0.99$, indicating that buses within the same zone respond near-identically to network disturbances. Inter-zone correlations (off-diagonal) are lower, particularly involving Zone 3 ($r = 0.884$ with Zone 2, $r = 0.911$ with Zone 4), whose lateral branch topology produces voltage dynamics partially decoupled from the main feeder. This spatial heterogeneity in coupling strength motivates the use of inter-zone electrical admittance as aggregation weights in the proposed INTACT framework. Zones that are more strongly coupled electrically should exert proportionally greater influence on each other's learned representations, a relationship that standard FedAvg ignores by treating all clients as equally weighted peers.

D. Stealthy Replay Attack Design

In a naive replay attack, an attacker substitutes real-time measurements with historical data via a hard cut. However, this introduces sudden temporal changes that localized time-series anomaly detectors, such as LSTM autoencoders, can easily identify. To rigorously evaluate the INTACT framework, we design a stealthy context-aware replay attack that remains consistent with local grid operating conditions while exploiting the spatial blind spot of zone-isolated detection.

Let the cyber-physical state of a zone at time t be the complete multi-variable measurement vector:

$$y(t) = [P(t), Q(t), |V(t)|, \theta(t)]^T \quad (1)$$

All four physical state variables including active power P , reactive power Q , voltage magnitude $|V|$, and phase angle θ , are simultaneously replaced at every timestep of the attack window. Full multi-variable substitution preserves the internal AC power flow consistency of the attacked zone: because all four variables are jointly replaced with historically coherent values, the local LSTM reconstructs the injected sequence with low residual $e(t) = \|y(t) - \hat{y}(t)\|^2 \leq \tau_{th}$, preventing detection by the isolated edge node.

The attack injection is carried out in two design steps:

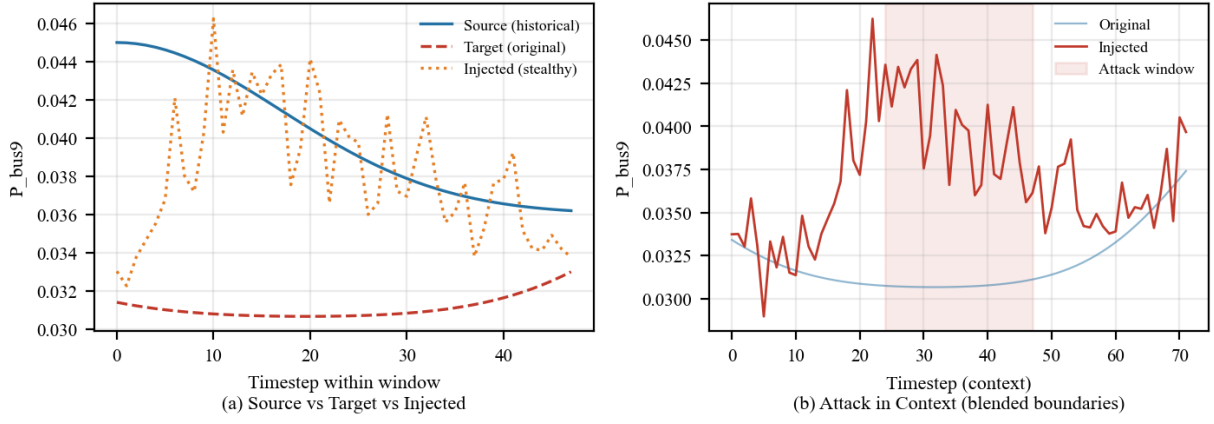


Fig. 3: Stealthy replay attack design for Zone 2. (a) Statistical window selection. (b) Cross-fade boundary blending.

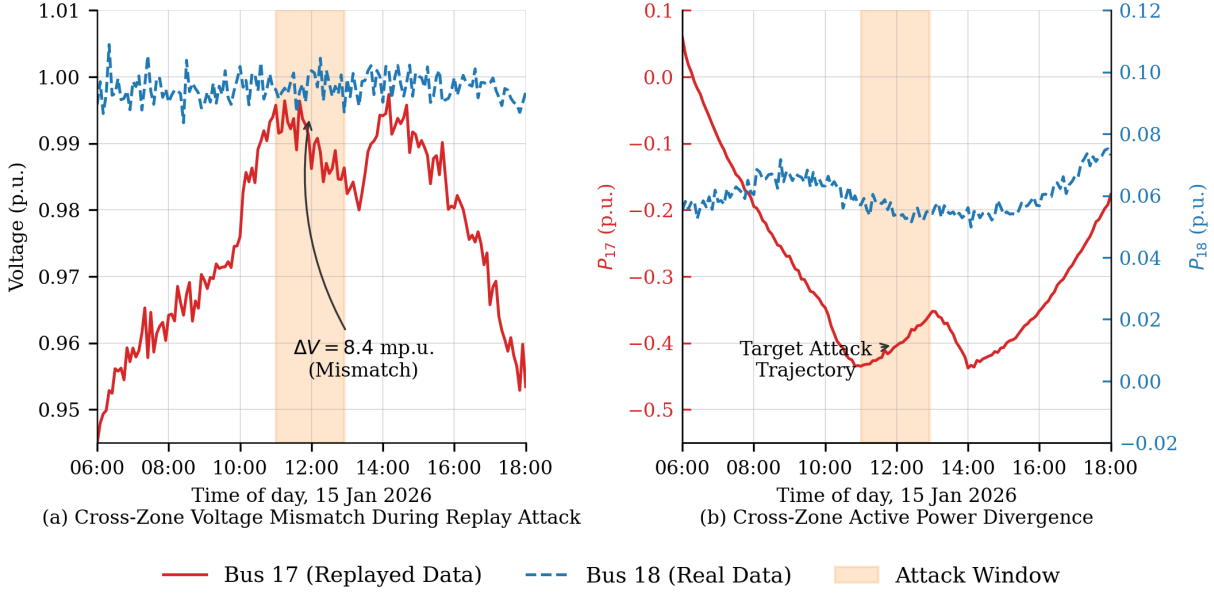


Fig. 4: Cross-zone spatial mismatch between compromised Bus 17 and unaffected Bus 18.

- **Statistical Window Selection:** Rather than replaying arbitrary historical data, the attacker identifies a past time window whose statistical properties such as mean and standard deviation across all features closely match the target window [19]. For DER-equipped zones (Zones 2–4), targets are peak solar generation periods (10:00–14:00). Source windows are selected at least two hours away from the target time and at least three days earlier to avoid trivial matching in the synthetic dataset.
- **Cross-Fade Boundary Blending:** To mask the injection boundary, sensor readings are cross-faded over 12 timesteps (60 minutes) at each injection edge, smoothly transitioning from real-time to historical values and back [20]. Only the unblended middle portion is labeled as an attack event. This eliminates the sharp temporal discontinuity to which LSTM-based detectors

are most sensitive [21]. Fig. 3 illustrates the result: Panel (a) shows that the source window shares a statistically similar operational range to the target, and Panel (b) shows the blended injection embedded in surrounding normal data with no detectable boundary.

E. Cross-Zone Spatial Inconsistency

Although these design steps make the attack locally undetectable, they introduce a spatial inconsistency that INTACT is designed to expose. Replacing all measurements in one zone with data from a different historical period breaks the physical coupling between neighboring zones. As shown in Fig. 4, during a Zone 2 attack at 11:00, Bus 17 reports replayed DER output with active power injection near -0.40 p.u., while adjacent Bus 18 in Zone 3 continues reflecting the true operating state with stable voltage near 1.00 p.u. In

a real grid, such a large DER shift at Bus 17 would also affect Bus 18 through feeder coupling. Since no corresponding change occurs, the resulting voltage mismatch of 8.4 mp.u. becomes physically impossible and reveals the replay attack. This inconsistency is invisible to the isolated local model but detectable through the INTACT cross zone consistency check.

F. Evaluation Dataset and Attack Injection Strategy

Stealthy replay attacks are injected independently into the test data of Zones 2–4, with four attack events per zone. Each injection spans a total of 48 timesteps (4 hours at 5-minute resolution), comprising 12-timestep cross-fade transitions on each boundary and a 24-timestep (2-hour) unblended core that is labeled as an attack event. For local evaluation, each model is tested only on its own attacked zone data containing the corresponding 28–36 bus level features.

For centralized and FL evaluation, the test data from all four zones are merged into a single dataset with 128 total features. Since Zone 1 does not contain DER infrastructure, replay attacks in that zone cannot use solar generation variability as a concealment mechanism and therefore fall outside the DER focused threat model considered in this work. As a result, Zone 1 remains unattacked during evaluation and serves as a normal reference region within the combined dataset.

The global attack label is set to 1 whenever any DER zone is under attack, resulting in 472 attack windows out of 4,002 total windows (11.8%). This evaluation setup tests whether centralized and FL models can capture cross zone spatial relationships that are not visible to isolated local models.

IV. ISOLATED AND CENTRALIZED DETECTION BASELINES

Two baseline architectures are evaluated to benchmark the INTACT framework: an isolated local LSTM Autoencoder, where each model trains only on data from its own zone, and a centralized LSTM Autoencoder that represents the theoretical upper bound with full data access. For both setups, input sequences span $T = 30$ consecutive timesteps (2.5 hours) covering four physical state variables per bus: P , Q , $|V|$, θ .

A. LSTM Autoencoder Architecture

Each zone uses an LSTM Autoencoder for anomaly detection. LSTMs are well suited for time-series power system data because they can learn temporal patterns from sequential measurements [22]. The autoencoder is trained on normal operating data and learns to reconstruct typical grid behavior. During inference, abnormal events such as replay attacks produce larger reconstruction errors, which are used as the anomaly score for detection.

The encoder compresses the input sequence through two stacked LSTM layers consisting of 32 units and then 64 units. This produces a compact latent representation of the zone's temporal operating pattern. The decoder mirrors this structure with 64 units followed by 32 units, reconstructing the original sequence through a time distributed Dense output layer.

Reconstruction quality is measured by the per window mean squared error:

$$e(t) = \frac{1}{T \cdot F} \sum_{\tau=1}^T \|y_i(t + \tau) - \hat{y}_i(t + \tau)\|_2^2 \quad (2)$$

where $T = 30$ is the window length and F is the number of input features per timestep. This includes 28–36 features per zone (4 features $\times$ 7–9 buses depending on zone).

The LSTM autoencoder architecture was kept fixed across all experiments to ensure a fair comparison. Using a common detection model allows the impact of different federated aggregation strategies to be evaluated independently of the underlying anomaly detector. For federated and centralized setups, learning rate and dropout were tuned on the combined dataset via Optuna [23]. Local baseline models were independently tuned per zone as described in Section IV-C. Finally, the detection threshold τ_{th} is calibrated per model at a high percentile of the training reconstruction errors, with the exact percentile selected by the local noise profile.

B. Edge Computing Testbed

All experiments run on a physical testbed of five NVIDIA Jetson devices [24]. This setup matches the actual hardware limits and computing constraints found in a real substation.

One NVIDIA Jetson Nano is assigned to each of the four zones to act as the local edge client. Each Nano has 4 GB of RAM and a 128 core Maxwell GPU. The central server runs on an NVIDIA Jetson AGX Xavier, which has 16 GB of RAM, a 512 core Volta GPU, and 64 Tensor Cores. This Xavier handles the INTACT aggregation logic and performs the admittance weighted mixing after each round.

The five edge devices communicate over a local Ethernet network. The Flower (flwr) FL framework handles the exchange of model weights [25]. Most importantly, the raw operational data never leaves the local Jetson Nanos.

C. Decentralized Local LSTM Autoencoder Training

The local configuration trains one LSTM Autoencoder per zone, utilizing exclusively that zone's measurement data. Since zones share no information during training or inference, this setup mimics edge nodes operating as completely isolated anomaly detectors.

To ensure a mathematically fair baseline, each model was optimized independently using the Optuna framework. This process allowed every local model to converge on its optimal architecture, including specific hidden unit counts and learning rates, tailored specifically to its unique DER variance. Subsequently, independent anomaly detection thresholds (τ_{th}) were calculated for each zone. These thresholds were established at a high percentile of the local training reconstruction errors, ranging from the 99.1st to the 99.9th percentile depending on the specific local noise profile.

During inference, any reconstruction error exceeding the localized τ_{th} boundary triggers an anomaly alarm. Table III details the detection performance of each isolated zone against the stealthy replay attack.

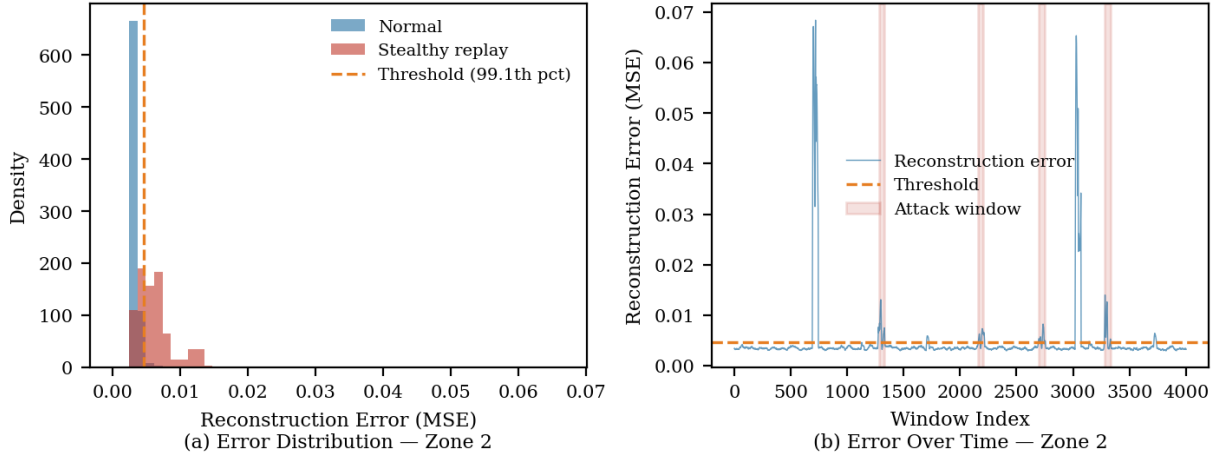


Fig. 5: Localized LSTM Autoencoder performance on Zone 2.

TABLE III: Localized Baseline Detection Performance

Zone	Thresh.	Prec.	Recall	F1	AUC	FPR
Zone 2	0.0045	0.44	0.69	0.54	0.903	0.049
Zone 3	0.0059	0.74	0.75	0.74	0.923	0.015
Zone 4	0.0033	0.68	0.94	0.79	0.985	0.025
Avg	—	0.62	0.79	0.69	0.937	0.030

The localized detection performance varies significantly across zones, directly reflecting the heterogeneous operating conditions induced by DER integration. Zone 4, which hosts the largest DER (0.6 MW at Bus 32), achieves a system-high recall of 0.94 because the suppression of such a massive active power signal creates a severe statistical discontinuity. In contrast, Zone 2 (0.5 MW at Bus 17) struggles with an F1-Score of only 0.54. Rather than creating a clear anomaly, the natural voltage volatility of the Zone 2 solar panel artificially elevates the local reconstruction noise floor, effectively masking the stealthy replay attack within normal operational variance.

Fig. 5 visualizes the local LSTM performance for Zone 2, with Panel (a) illustrating a massive overlap between normal and attack reconstruction errors where the attack windows barely push past the detection threshold. Panel (b) confirms this timeline by showing that three of the four attack events generate reconstruction errors that only marginally exceed the threshold. Furthermore, a false positive spike occurs near window 720, reflecting a normal operational change such as natural variations in solar generation or load demand that the model incorrectly classified as an anomaly. These results highlight a major architectural flaw, confirming that even perfectly tuned isolated local detectors cannot reliably catch stealthy replays hidden inside natural DER variability.

D. Centralized LSTM Autoencoder Training

The centralized baseline takes a different approach. It trains a single LSTM Autoencoder on the combined raw data from all four zones at once. The input space spans 128 features at

each timestep. This gives the model complete visibility over the entire physical system.

We used Optuna to tune the hyperparameters on the combined dataset. The best configuration consists of two LSTM layers (32 and 64 units) with no dropout and a learning rate of 5.73×10^{-3} . The detection threshold was set at the 99.5th percentile of the training reconstruction errors ($\tau_{th} = 0.004705$). To ensure a fair comparison, this exact architecture was kept for all FL experiments.

The centralized model achieved an F1-Score of 0.8 and an AUC-ROC of 0.971, significantly outperforming the local models (average F1-Score of 0.69). This confirms the hypothesis from Section III: seeing the whole grid at once is required to catch stealthy replay attacks. The centralized model detects spatial mismatches that simply look like normal solar fluctuations to an isolated local model.

However, this architecture is impractical for real-world deployment. Sending raw telemetry to a central cloud violates NERC CIP-005 and CIP-011 cybersecurity and information-protection requirements [26], [27]. This regulatory roadblock is precisely why a federated approach is necessary. The INTACT framework, introduced in Section V, is designed to achieve this same cross-zone awareness without ever moving raw data off the edge devices.

V. PROPOSED TOPOLOGY-AWARE FEDERATED LEARNING

The proposed INTACT framework operates in two phases: training and inference. The IEEE 33-bus system is partitioned into four zones, with DERs located at Buses 17, 24, and 32. During training, each Jetson Nano edge client trains a local LSTM autoencoder using only its zone’s data and shares model updates with a central Jetson AGX Xavier aggregator. The aggregator performs admittance-weighted personalized aggregation (Eqs. 4–6) and returns a customized model to each zone without exchanging raw data. During inference, each zone computes a local reconstruction error (Eq. 7), which is compared with a neighbour-weighted spatial average (Eq. 8)

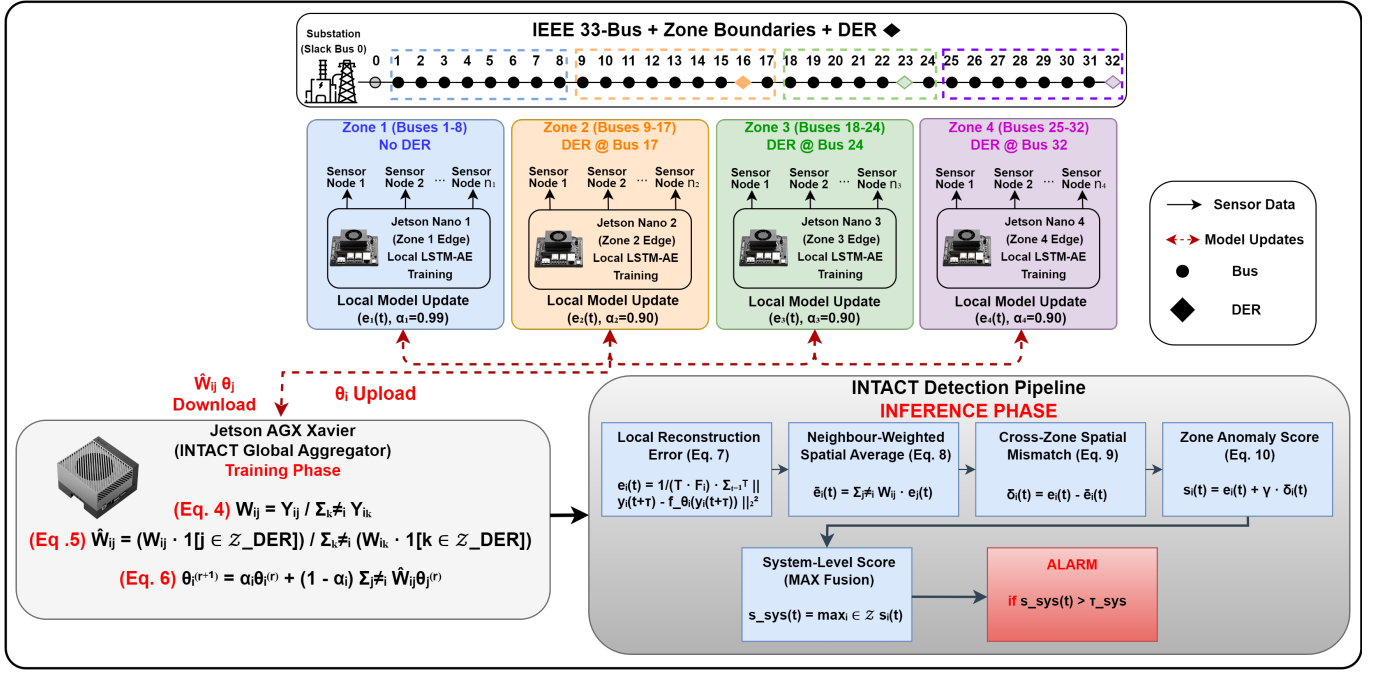


Fig. 6: INTACT framework for topology-aware federated anomaly detection.

to measure cross-zone inconsistency (Eq. 9). These scores are combined to produce a zone-level anomaly score (Eq. 10), and the maximum score across all zones is used to generate the final system-level alarm (Eqs. 11–12). The overall architecture is illustrated in Fig. 6.

A. Limitations of Standard Federated Aggregation

In standard FedAvg [8], the server aggregates client updates as a sample weighted average:

$$\theta^{(r+1)} = \sum_{i=1}^N \frac{n_i}{n} \theta_i^{(r)} \quad (3)$$

This approach treats all clients as statistically equal. It assigns influence based purely on dataset size and ignores the physical connections between zones. As shown in Section III-C, the physical coupling in the IEEE 33-bus network is highly uneven. For example, Zone 3 has a Pearson voltage correlation of only $r = 0.884$ with Zone 2 because of its laterally branched layout. Meanwhile, correlations within a single zone consistently stay above $r = 0.99$.

FedAvg completely ignores this physical structure. Under this method, a pair of tightly connected zones receives the exact same shared model influence as two zones that are physically distant. Furthermore, FedAvg forces every client to use the exact same global model. Each zone receives a blend of weights from completely unrelated areas. This blending washes out the specific local data patterns needed to actually detect localized anomalies.

The results clearly show this flaw. FedAvg only achieves an F1-Score of 0.52, falling 28 percentage points short of the centralized upper bound. The INTACT framework addresses

both problems: first by weighting inter-zone model influence according to physical admittance, and second by explicitly checking cross-zone reconstruction consistency at inference time.

B. Admittance-Weighted Aggregation (INTACT)

The electrical coupling between zones depends on the total admittance of all transmission lines crossing their boundary. We define Y_{ij} as the sum of physical admittance between connected buses in zone i and zone j . Using these values, the row normalized aggregation weights are defined as:

$$W_{ij} = \frac{Y_{ij}}{\sum_{k \neq i} Y_{ik}} \quad (4)$$

Zone 1 does not contain any DER infrastructure. Therefore its normal load changes are completely separated from the solar driven dynamics of the other zones. If Zone 1 contributed equally to the other models during aggregation it would weaken the important physical relationships that the INTACT framework relies on. To prevent this we compute a special weight $\hat{W}_{ij}$ by setting the weight of neighbors without solar power to zero and renormalizing:

$$\hat{W}_{ij} = \frac{W_{ij} \cdot \mathbf{1}[j \in \mathcal{Z}_{\text{DER}}]}{\sum_{k \neq i} W_{ik} \cdot \mathbf{1}[k \in \mathcal{Z}_{\text{DER}}]} \quad (5)$$

The resulting weights for the IEEE 33-bus testbed are presented in Table IV. Zone 1 is fully excluded ($\hat{W}_{i1} = 0$ for all DER zones), Zones 2 and 3 each draw 80% of their cross-zone influence from Zone 4 due to its stronger tie-line admittance, and Zone 4 splits equally between Zones 2 and 3 since both tie-lines carry identical physical admittance.

TABLE IV: DER-Aware Aggregation Weights $\hat{W}_{ij}$ for the IEEE 33-bus Testbed

Zone	Z1	Z2	Z3	Z4
Z1	—	0.094	0.544	0.362
Z2	0.000	—	0.200	0.800
Z3	0.000	0.200	—	0.800
Z4	0.000	0.500	0.500	—

Algorithm 1 INTACT Training Protocol

Require: Zones $\mathcal{Z} = \{z_1, z_2, z_3, z_4\}$, rounds $R = 20$, local epochs $E = 3$, admittance matrix, $\alpha = 0.90$ (active zones), $\alpha_1 = 0.99$ (Zone 1)

Ensure: Per zone personalized models $\{\theta_1, \theta_2, \theta_3, \theta_4\}$

- 1: Compute base admittance Y_{ij} from zone data
 - 2: Compute normalized weights W_{ij} via Eq. (4)
 - 3: Compute specialized weights $\hat{W}_{ij}$ via Eq. (5)
 - 4: Initialize all $\theta_i \leftarrow \theta_0$
 - 5: **for** $r = 1$ to R **do**
 - 6: Server dispatches personalized $\theta_i^{(r)}$ to zone i
 - 7: Zone i trains locally for E epochs
 - 8: Zone i returns updated $\theta_i^{(r)}$ to server
 - 9: Server computes mixed $\theta_i^{(r+1)}$ via Eq. (6) for all i
 - 10: **end for**
 - 11: **return** $\{\theta_1, \theta_2, \theta_3, \theta_4\}$
-

This modified weight is used exclusively during training. The original physical weight W is kept for the consistency check during inference to preserve the actual grid topology.

Unlike standard federated averaging, INTACT maintains a separate model θ_i for each zone. The server dispatches each specific model at the start of every round instead of a single shared global model. Each zone performs three local epochs to minimize its mean squared reconstruction loss. After local training each zone returns its updated weights to the server. The server then performs an admittance weighted mixing step:

$$\theta_i^{(r+1)} = \alpha_i \theta_i^{(r)} + (1 - \alpha_i) \sum_{j \neq i} \hat{W}_{ij} \theta_j^{(r)} \quad (6)$$

The retention parameter α_i controls the balance between local specialization and cross zone knowledge transfer. The active solar zones use $\alpha_i = 0.90$. This retains 90 percent of their locally trained weights while absorbing 10 percent of the neighbor influence weighted by physical admittance. Zone 1 uses $\alpha_i = 0.99$. This keeps Zone 1 largely local, preventing passive dynamics from influencing DER-connected zones. The complete training procedure is summarized in Algorithm 1.

C. Cross-Zone Consistency Verification

The training process naturally embeds physical relationships into the model of each zone. During testing, INTACT adds an explicit consistency check. This check increases the anomaly score for any zone that has a higher reconstruction error than its physical neighbors expect.

For each test window at time t , we calculate the raw mean squared reconstruction error for zone i as:

$$e_i(t) = \frac{1}{T \cdot F_i} \sum_{\tau=1}^T \|\mathbf{y}_i(t + \tau) - f_{\theta_i}(\mathbf{y}_i(t + \tau))\|_2^2 \quad (7)$$

The physical admittance weights W_{ij} help calculate the expected error level based on the current conditions of neighboring zones:

$$\bar{e}_i(t) = \sum_{j \neq i} W_{ij} \cdot e_j(t) \quad (8)$$

We then calculate the spatial mismatch to see exactly how far the actual error deviates from this expected neighbor level:

$$\delta_i(t) = e_i(t) - \bar{e}_i(t) \quad (9)$$

Finally, the total anomaly score combines the raw error and the weighted mismatch:

$$s_i(t) = e_i(t) + \gamma \cdot \delta_i(t) \quad (10)$$

Here $\gamma = 0.3$ acts as the consistency penalty coefficient. Under normal conditions, the physical connections keep the reconstruction errors correlated across zones. This means $e_i(t) \approx \bar{e}_i(t)$, making the mismatch near zero.

During a replay attack, the injected data comes from a completely different historical time. This breaks the expected physical correlations. The error of the attacked zone rises compared to its healthy neighbors. This creates a positive mismatch and pushes the final score above the detection threshold. This catches the attack even if the raw error alone was too small to trigger an alarm in a standard blind system.

To produce a single system-level decision, the per-zone scores are fused by taking the maximum across all zones:

$$s_{\text{sys}}(t) = \max_{i \in \mathcal{Z}} s_i(t) \quad (11)$$

This MAX fusion ensures that a localized attack in any single zone is sufficient to trigger a global alarm, without requiring all zones to agree. A binary alarm is raised when this system score exceeds a threshold τ_{sys} :

$$\text{alarm}(t) = \begin{cases} 1 & \text{if } s_{\text{sys}}(t) > \tau_{\text{sys}} \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

The threshold τ_{sys} is calibrated as the 99.6th percentile of $s_{\text{sys}}(t)$ computed on clean training windows, ensuring a false positive rate below 0.4% on normal grid operation.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

All FL experiments used a strictly uniform training configuration to ensure a fair comparison. Each method trained for 20 communication rounds. Within each round, every zone client performed three local epochs on its own private data before returning its model updates to the central AGX Xavier server. All four zones actively participated in every single round. The batch size of 32, the learning rate of 5.73×10^{-3} , and the LSTM architecture were kept exactly identical across all methods to perfectly match the

TABLE V: Global Architecture Performance Comparison.

Architecture	Privacy	Topology-Aware	Precision	Recall	F1-Score	AUC	FPR	Inference Time (s)
Centralized LSTM (upper bound)	No	No	0.708	0.920	0.800	0.971	0.051	—
Local LSTMs Avg (Zones 2/3/4)	Yes	No	0.620	0.793	0.690	0.937	0.030	—
FedAvg	Yes	No	0.548	0.494	0.520	0.915	0.054	11.19
FedProx ($\mu=0.001$)	Yes	No	0.508	0.477	0.492	0.909	0.062	11.24
FedAdam ($\eta=0.01$)	Yes	No	0.509	0.663	0.576	0.892	0.086	10.94
Ditto ($\lambda=0.1$)	Yes	No	0.655	0.439	0.525	0.919	0.031	11.62
INTACT (Proposed)	Yes	Yes	0.664	0.811	0.730	0.930	0.055	11.12

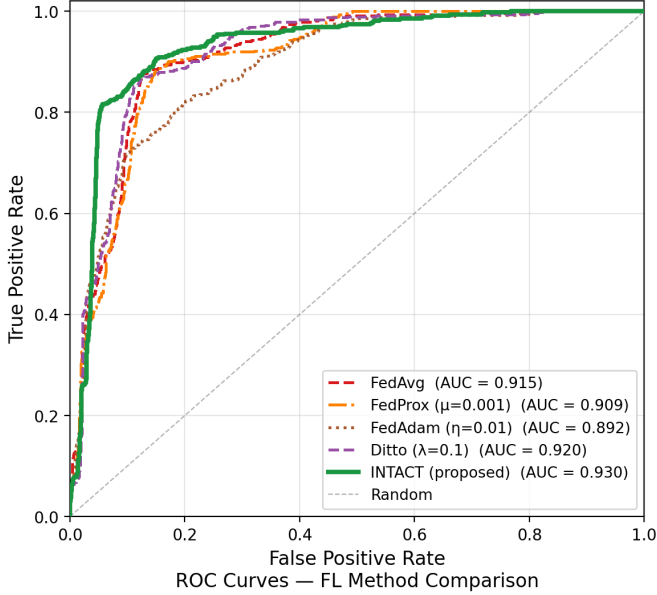


Fig. 7: ROC curves of all FL methods on the 4,002-window testing dataset.

centralized baseline described in Section IV-D. Finally, the method specific hyperparameters were selected using a held out validation sweep. These optimized values included $\mu = 0.001$ for FedProx, $\eta = 0.01$ for FedAdam, $\lambda = 0.1$ for Ditto, alongside $\alpha = 0.90$ and $\gamma = 0.3$ for the proposed INTACT framework.

A. FL Architecture Comparison

As shown in Table III, the average local F1-score of 0.690 is driven up by Zone 4 (0.79), where the strong 0.6 MW solar unit makes anomalies easier to detect. Zone 2 is more challenging, with an F1-score of 0.54 due to higher variability. Since these models operate independently, attacks in one zone are not visible to others, highlighting the need for FL for system-wide detection while preserving data privacy. Table V summarizes overall performance, and Fig. 7 shows the ROC curves for the FL methods.

1) **Topology-blind FL baselines:** Standard FL baselines, including FedAvg [8], FedProx [28] ($\mu = 0.001$), and FedAdam [29] ($\eta = 0.01$), all demonstrate similar overall behavior. They achieve F1-scores of 0.520, 0.492, and 0.576 respectively, with AUC values ranging between 0.892 and

0.915. Even though these methods share updates across zones, they do not use any information about how the grid is physically connected. Their aggregation is based purely on dataset size, completely ignoring the physical coupling structure of the grid. As a result, the global model learns an averaged representation that is not well aligned with the solar behavior of any individual zone. This limits its ability to capture the spatial patterns created by replay attacks. FedAdam achieves the highest recall at 0.663 due to its adaptive learning rate strategy, but its precision drops to 0.509. This indicates that while it detects more attacks, it also produces a much higher number of false alarms.

2) **Personalized FL (Ditto):** Personalized FL techniques such as Ditto [30] ($\lambda = 0.1$) achieve the highest precision among all tested methods at 0.655. However, it also records the lowest recall at 0.439. This results in an F1-Score of 0.525, which is nearly identical to the standard FedAvg. This failure is a direct result of the attack design. The attacker uses statistical window selection and boundary blending to make the injected data look perfectly natural within the historical baseline of that specific zone. This vulnerability is directly amplified by Ditto’s two-phase training architecture. Unlike standard FL methods that use only the shared global model, Ditto first completes a full FedAvg training phase (20 rounds $\times$ 3 local epochs) and then runs a separate 60-epoch personalization phase per zone. In this phase, each zone minimizes a local MSE objective regularized by a proximal term that keeps the personalized model close to the FedAvg anchor. The personalization budget of 60 epochs was deliberately matched to the FL training budget to ensure a fair computational comparison. This intensive zone-specific adaptation makes each model an expert at reconstructing its own normal profile. Because the replayed attack data perfectly mimics this normal profile, the local models easily reconstruct the injected signal. The reconstruction error stays very low during the attack, allowing the manipulation to slip past the high precision threshold. Although Ditto achieves an AUC of 0.920, lowering the threshold to detect these attacks would significantly increase false alarms.

3) **INTACT:** The proposed INTACT framework achieves the highest performance across all tested FL methods. It records an F1-Score of 0.730, a recall of 0.811, a precision of 0.664, and an AUC of 0.930. When compared to the centralized upper bound ($F1 = 0.800$, recall = 0.920), INTACT closes 75 percent of the gap between FedAvg and

the centralized upper bound while completely preserving data privacy across all zones. Furthermore, it improves the F1-score by 15.4 percentage points compared to FedAdam.

The ROC curve in Fig. 7 confirms this distinct advantage. The curve for INTACT lies strictly above all baselines at low false positive rates ($FPR < 0.15$). The FPR values in Table V reflect the operating point of each method at its chosen threshold. Furthermore, the ROC curve shows that at any fixed FPR, INTACT achieves substantially higher recall. For example, at an $FPR \approx 0.055$, INTACT reaches a recall of 0.811 while FedAvg reaches only 0.494. This specific operating region is critical for real world power systems where false alarms trigger expensive manual inspections. Finally, the inference latency is 11.12 seconds per test window. This is practically identical to FedAvg (11.19 s), showing that physical topology awareness and cross-zone consistency checks introduce no measurable overhead.

B. Spatial Mismatch Analysis

The replay attack is specifically engineered to bypass local anomaly detectors. Every physical variable (P , Q , $|V|$, θ) in the compromised zone is replaced with historical data that statistically resembles the current operating point. Consequently, the local LSTM reconstructs this injected sequence with very low error, allowing the attack to slip past the detection threshold completely unnoticed.

However, the attack introduces a physical contradiction that it cannot hide because the neighboring zones remain untampered. In a physical power grid, a significant load or generation change at a location like Bus 17 in Zone 2 naturally propagates through the feeder, causing a measurable voltage shift in the adjacent Zone 3. When this expected shift does not occur because Zone 3 continues to report its true unaltered measurements, a clear spatial mismatch emerges. This mismatch highlights a fundamental disagreement between what the compromised data in Zone 2 implies and what the healthy neighboring zones actually report.

The consistency score defined in Eq. 10 amplifies the overall anomaly score by applying the specific penalty coefficient:

$$s_i(t) = e_i(t) + 0.3 \cdot (e_i(t) - \bar{e}_i(t))$$

During an attack, the raw error $e_i(t)$ rises marginally in the compromised zone. Simultaneously, the admittance weighted average error of the healthy neighboring zones, denoted as $\bar{e}_i(t)$, remains low. The resulting gap between these two values is multiplied by the penalty coefficient $\gamma = 0.3$ and added to the raw score. This amplifies the final anomaly score, pushing it above the detection threshold even in cases where the raw error alone was insufficient to trigger an alarm.

The personalized FL baseline, Ditto, fails precisely at this juncture. Its intensive local fine tuning process makes each model an extreme expert at reconstructing its own data distribution. Unfortunately, this includes the replayed attack data that was deliberately crafted to mimic normal operations. Because the overfitted local model perfectly reconstructs the injected signal, $e_i(t)$ remains flat during the attack. This

completely suppresses the spatial mismatch signal and allows the manipulation to pass through. This dynamic explains why Ditto achieves the highest precision at 0.655 while simultaneously recording the lowest recall at 0.439 out of all tested methods. It can easily flag massive uncoordinated errors, but the locally coherent nature of the replay attack ensures those errors never become large enough to detect.

The controlled aggregation of INTACT prevents this over specialization. By utilizing a self retention parameter of $\alpha = 0.90$ as detailed in Section V-B, the framework forces each zone to remain sensitive to spatially foreign patterns rather than perfectly memorizing its own local history. This specific combination of topology guided aggregation during training and explicit cross zone mismatch scoring during inference is exactly what allows INTACT to achieve a high recall of 0.811 while adding zero additional inference latency compared to standard FedAvg.

VII. CONCLUSION

This paper presented INTACT, a physically aware FL framework designed to detect stealthy replay attacks in modern power grids. The core insight is straightforward but highly effective. While these cyberattacks are cleverly disguised to look completely normal to an isolated local sensor, they unavoidably break the expected electrical relationships with neighboring zones. INTACT uses this physical mismatch to expose the attack.

The framework accomplishes this through two coordinated mechanisms. During training, the system uses the actual physical grid connections to guide how the models share knowledge. Zones that are tightly coupled electrically have a much stronger influence on each other. During testing, INTACT applies a consistency check. It flags an anomaly any time a zone reports data that severely disagrees with the expectations of its physical neighbors. It achieves all of this without ever transferring raw private data away from the local edge devices.

Testing on the IEEE 33-bus distribution system using NVIDIA Jetson hardware proved highly successful. INTACT achieved an F1-Score of 0.730 and a recall of 0.811. This brings the accuracy remarkably close to the centralized upper bound, proving that data privacy does not need to be sacrificed to achieve high detection performance. The experiments also revealed a surprising flaw in personalized FL. Because algorithms like Ditto focus so heavily on adapting to local data, they accidentally learn to accept the replayed attack data as normal. This caused Ditto to have the lowest recall of all tested methods despite having the highest precision. Finally, INTACT runs just as fast as standard methods. It completes inference in 11.12 seconds compared to 11.19 seconds, proving that adding physical awareness introduces zero measurable runtime delay.

Future work will scale the INTACT framework to much larger systems, specifically testing its performance on the IEEE 123-bus network. Additional research will also evaluate its robustness against coordinated attacks targeting multiple

zones simultaneously and explore new ways to reduce communication overhead between the edge devices.

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